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Semester-V

Computer Graphics and Multimedia

Paper Code: BCAMACO501T22

Course outcomes:
At the completion of the course, a student will be able to:

1. Remember the technical aspect of Multimedia Systems.
2. Understand various file formats for audio, video and text media.
3. Develop various Multimedia Systems applicable in real time.
4. Design interactive multimedia software.
5. Apply various networking protocols for multimedia applications.
6. Evaluate multimedia application for its optimum performance.

UNIT-I

Introduction: The Advantages of Interactive Graphics, Representative Uses of Computer Graphics, Classification of Application Development of Hardware and software for Computer Graphics, Conceptual Framework for Interactive Graphics, Overview, Scan: Converting Lines, Scan Converting Circles, Scan Converting Ellipses.

UNIT-II

Hardcopy Technologies, Display Technologies, Raster-Scan Display System, Video Controller, Random-Scan Display processor, Input Devices for Operator Interaction, Image Scanners. Clipping Southland - Cohen Algorithm, Cyrus-Beck Algorithm, Midpoint Subdivision Algorithm.

UNIT-III

Geometrical Transformation: 2D Transformations, Homogeneous Coordinates and Matrix Representation of 2D Transformations, Composition of 2D Transformations, The Window-to-Viewport Transformation, Efficiency, Matrix Representation of 3D Transformations, Transformations as a Change in Coordinate System.

UNIT-IV

Multimedia Elements, Multimedia Applications, Multimedia System Architecture, Evolving Technologies for Multimedia Systems, Multimedia Data Interface Standards, the Need for Data Compressions, Multimedia Database.

UNIT-V

Media and Data Streams: Medium, Main Properties of a Multimedia Stream, Multimedia System Definition, Combination of Media. Data & File Format Standards: Rich –Text Format, TIFF File Format, RIFF, MIDI File Format, JPEG DIB File Format, MPEG
Standards.

Text Books:

1. Foley, Van Dam, Feiner, Hughes, Computer Graphics Principles& practice,2000.
2. Ralf Skinmeiz and KlanaNaharstedt, Multimedia: computing, Communication and Applications, pearson, 2001
3. D.Harn& Baker. Computer Graphics Prentice Hall of India,1986.

4. J. F. KoegelBuferd -Multimedia Systems, Pearson Education, New Delhi, 2006

Text Books:

1. Villamil and Molina, "An Introduction to Multimedia", MacMillan, 1997
2. Lozano, "Multimedia: Sound & Video", Prentice Hall of India (Que), 1997.
3. Ranjan Parekh, "Principle of Multimedia", Tata McGraw Hilf.
4. Villamil and Molina, "Multimedia: Production, Planning and Delivery", Que, 1997.

Reference Books:

5. Sinclair, "Multimedia on the PC". BPB Publications
6. Tay Vaughan. "Multimedia: Making It Work", Fifth edition, Tata McGraw Hill, 1994.
7. James E Shuman, "Multimedia in Action", Wadsworth Publications, 1997.
8. Jeff Coate Judith, "Multimedia in Practice", Prentice Hall of India, 1995 John F. Koegel, "Multimedia Systems", Addison Wesley Ltd.